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Evaluation of electronic medical records risk management in Kudus hospitals: A comparative qualitative study

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ABSTRACT

Background: The rapid expansion of digital health infrastructure has positioned electronic medical records (EMR) as a core element of service continuity, clinical decision-making, and data governance, yet the maturity of EMR risk management in many developing health systems remains uneven and weakly institutionalized.
Aims: This study evaluates and compares EMR risk management practices in two Indonesian hospitals, mapping how organizational capability, system architecture, and regulatory context shape maturity progression and digital resilience.
Methods: A comparative qualitative design was applied using semi-structured interviews, focus groups, non-participant observation, and document review in a regional public hospital (DK) and a private hospital (AK). Data were thematically coded and triangulated, then interpreted through the HIMSS EMRAM maturity model, PDCA cycle, and SWOT analysis to assess governance readiness and risk-mitigation capability.
Results: DK Hospital demonstrated more advanced EMR maturity (transitioning toward EMRAM Stage 5 with approximately 50% paperless workflows), structured PDCA-based monitoring, and a formalized risk taxonomy spanning technical, human, regulatory, and operational domains. AK Hospital adopted a modular EMR development strategy that increased flexibility but generated integration gaps, documentation weaknesses, and stronger dependency on key IT personnel. Both hospitals shared critical vulnerabilities related to human error, hybrid paper–electronic environments, fragmented incident documentation, and persistent uncertainty around electronic signature regulation.
Conclusions: The findings show that EMR risk management maturity is determined less by technology acquisition than by multi-level governance, continuous monitoring loops, and the ability to manage uncertainty in complex socio-technical systems. By integrating EMRAM, PDCA, and SWOT perspectives, this study proposes an evidence-based roadmap for strengthening EMR risk governance in Indonesian hospitals and provides transferable insights for digital hospital resilience in other low- and middle-income health systems.

Introduction

Recent years have seen accelerated digital transformation of healthcare services, with electronic medical records (EMR) emerging as critical infrastructure for longitudinal patient data, clinical decision support, and institutional accountability. International evidence shows that well-designed EMR and electronic medication systems can reduce selected medication errors, improve documentation, and enable secondary use of data, although they also introduce technology-related, cyber, and medico-legal risks.[1–7] Consequently, EMR implementation is increasingly framed as the deployment of a complex socio-technical risk system rather than a narrow documentation upgrade.[4,8,9] In this sense, EMR implementation is better understood as the deployment of a complex socio-technical risk system rather than a purely technical documentation upgrade.

In Indonesia, Ministry of Health Regulation No. 24/2022 formalises a national commitment to EMR adoption and aligns domestic digitalisation with international data-governance principles.[10] Law No. 44/2009 on Hospitals and technical guidelines for risk management under the Secretariat General require internal control systems,

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systematic reporting, and patient-safety-oriented governance.[11–13] These instruments position EMR as a regulated risk object in which data security, service continuity, and conformity with national health insurance (BPJS) rules must be assured simultaneously.[9,14,15] Empirical work nonetheless suggests that implementation capacity and risk-management maturity remain heterogeneous across regions and ownership types [16-18] .

EMR studies in Indonesia and comparable settings have mainly examined adoption status, workflow efficiency, and perceived service quality, with limited analysis of formal risk-management routines.[8,12,19] Research in Central Java indicates that infrastructure readiness and staff training are critical for successful EMR rollout, whereas budget constraints, interoperability gaps, and user resistance remain persistent barriers. [19,20] International reviews confirm that electronic records can improve documentation completeness and enable data-driven decision-making, but benefits depend on robust governance structures and embedded risk-management processes.[3–5]

Parallel literature on hospital risk management documents uneven institutionalisation of enterprise risk management, with many hospitals confining formal risk activities to accreditation cycles or narrow clinical domains.[15,21–23] Studies in European and Asian hospitals show that communication gaps, limited staff engagement, fragmented incident reporting, and weak feedback loops hinder maturation of clinical risk-management systems. [2,21] However, EMR implementation and hospital risk governance are seldom analysed together, leaving a fragmented understanding of how digital infrastructures are embedded within broader organisational risk-management architectures. [17,22]

Power-systems and microgrid research offers mature theoretical and methodological tools for managing uncertainty, decentralisation, and diagnostics in safety-critical infrastructures.[24] Studies on islanded microgrids propose decentralised control architectures that coordinate distributed generation, energy storage, and load shedding to maintain stability under fluctuating renewable outputs and islanding, emphasising local autonomy and adaptive reconfiguration.[25] Smart-scheduling work integrates power management, voltage control, and distributed optimisation in unified frameworks that jointly manage cost, reliability, and resilience under prediction errors.[24]

Transformer-diagnostics research similarly treats uncertainty in complex signal patterns as a central design problem. Baroumand et al. [26] combine discrete wavelet transforms, logistic regression, and neural networks to classify winding faults and quantify the contribution of each wavelet component.[26] Parkash et al. [27] introduce a hyperbolic fuzzy cross-entropy methodology to interpret ambiguous frequency-response signatures and to discriminate between healthy and faulty transformer states across frequency bands.[27] Collectively, these studies conceptualise critical infrastructures as distributed, data-intensive systems in which uncertainty is managed through multi-level control, feature-rich diagnostics, and probabilistic reasoning rather than deterministic rule-sets.[26]

Translating these insights to hospitals suggests that EMR risk management should be framed as an uncertainty-rich, distributed control problem, where clinical units and IT services function analogously to microgrid nodes and incident logs, audit trails, and usage analytics mirror high-dimensional sensor data.[15,23,27] This perspective points to the need for integrated, data-driven risk-governance architectures, rather than ad hoc or accreditation-driven routines, to sustain EMR resilience.[4,8,9,15]

Despite increased attention to EMR adoption and hospital accreditation in Indonesia, there is still no study that systematically examines how risk-management principles are concretely applied within EMR implementation or how maturity models, continuous-improvement cycles, and institutional risk taxonomies interact in practice. [16,28,17] Existing EMR work largely stops at adoption outcomes or service-quality indicators, while risk-management studies rarely disaggregate digital-system risks from broader clinical or organisational risks.[2,19,21,22] Furthermore, theoretical tools developed in other safety-critical infrastructures for modelling uncertainty and distributed control have not yet been mobilised to interpret EMR risk governance.[24,27]

Against this backdrop, the present study examines how EMR-related risks are identified, managed, and governed in two hospitals in Kudus Regency (DK Hospital, a fully accredited regional public hospital and AK Hospital, a private hospital with developing EMR infrastructure). Using a comparative qualitative design that integrates HIMSS EMRAM, SWOT, and PDCA frameworks, the study maps differences in maturity trajectories, governance practices, and vulnerability profiles across technical, human, regulatory, and operational domains.

This study offers three contributions. First, it provides one of the first in-depth comparative analyses of EMR risk-management practices in Indonesian hospitals, linking maturity levels, PDCA implementation, and detailed risk taxonomies across two contrasting institutional contexts. Second, it imports insights from uncertainty and distributed-control theory in microgrids and transformer diagnostics to conceptualise EMR as a complex socio-technical risk system, enriching the framing of digital-hospital resilience. Third, it proposes an integrated roadmap for strengthening EMR risk governance that aligns EMRAM stages with ISO 31000 and COBIT 5 principles, tailored to Indonesian regulatory and infrastructural realities.

Stakeholder perspectives such as covering clinical staff, IT personnel, and hospital leadership are incorporated to gain a multidimensional understanding of effectiveness, obstacles, and adaptive mechanisms. Ultimately, this study is expected to contribute both conceptually and practically to the development of safer, more efficient, and patient-centered EMR implementation strategies in Indonesia.

Research methods

Analytical framework and risk domains

This study used a comparative qualitative design to examine EMR risk-management practices in two hospitals in Kudus, Central Java, Indonesia. DK Hospital is a fully accredited regional public hospital, whereas AK Hospital is a private hospital with a developing EMR system. The contrasting accreditation status and EMR maturity provided a deliberate basis for comparison within a shared regional and regulatory context.

Analysis focused on five pre-specified EMR risk domains: technical/system, security/privacy, human error, regulatory/compliance, and operational risks. These domains were used to structure within-case analysis and cross-case comparison, and to link empirical patterns with the broader theoretical lens of uncertainty and distributed control.

EMR maturity was assessed using the HIMSS EMRAM model, focusing on integration of laboratory, radiology, and pharmacy services, coverage of clinical documentation, CPOE and decision-support capabilities, and progress toward paperless workflows. SWOT analysis summarised strengths, weaknesses, opportunities, and threats related to EMR risk governance, while the Plan–Do–Check–Act (PDCA) cycle was used to characterise the depth of continuous-improvement routines.

Data collection and analysis

Data were collected through semi-structured interviews, focus group discussions (FGDs), non-participant observation, and document review.

Interviews and FGDs explored perceptions of EMR implementation, risk identification and response, and organisational readiness. Observations focused on real-time EMR use, including work-arounds and incident handling. Documents provided corroborating evidence on formal rules, reporting structures, and historical incidents. Written informed consent was obtained from all participants, and personal identifiers were removed from transcripts.

Transcripts, field notes, and documents were coded using a framework that combined the five risk domains with EMRAM, SWOT, and PDCA categories. Initial codes were derived deductively from these

categories and then refined inductively as new themes emerged. Within-case analyses were conducted first to build detailed narratives for DK and AK Hospitals, followed by cross-case matrices to compare maturity levels, governance practices, and vulnerability profiles.

Credibility was supported by triangulation across methods and data sources, peer debriefing within the research team, member checking with selected participants, and maintenance of an audit trail of coding and analytic decisions.

Handling of equations and quantitative indicators

Although the study is mainly qualitative, some formal expressions are used for clarity, such as generic risk scores, maturity indices, and percentage-based indicators. For each equation, all symbols are defined immediately in the text, even if they also appear in the nomenclature for example $R=P \times I$, (for instance, R = overall EMR risk score, P = probability of occurrence, I = impact on hospital operations). Subscripts and superscripts (such as R_{tech} and R_{human}) are defined at first use, and notation is kept consistent across the manuscript. For percentages and ratios, the denominator is explicitly stated ("percentage of total medical record forms" or "IT personnel per 100 licensed beds") to ensure unambiguous interpretation.

Results

EMR implementation trajectories in both hospitals

DK Hospital began EMR implementation in June 2022 in ambulatory care, followed by phased rollout in inpatient and emergency departments during 2024. At the time of data collection, approximately 50% of medical record forms were fully electronic, with the remaining documentation (particularly in inpatient units) still maintained in hybrid paper–electronic format. Core electronic modules included initial assessment, integrated progress notes, discharge summaries, pharmacy records, diagnostic reports, consultations, and operative notes, allowing partial end to end digital clinical workflows.

AK Hospital initiated EMR deployment in late 2022 for ambulatory care using a system provided through the PKOGAP programme (Aisyiyah Creativity and Mutual Cooperation Program) of Universitas Gadjah Mada, Yogyakarta. They subsequently started migrating inpatient documentation in 2024 to an internally developed platform. The modular development strategy provided flexibility to tailor functionalities to local needs but generated complex integration challenges, particularly in consolidating multiple legacy databases and aligning data structures across subsystems. The decision to internalise system development was driven by dissatisfaction with vendor solutions that were perceived as poorly structured and misaligned with hospital operations.

HIMSS EMRAM maturity model framework analysis

EMR maturity assessed using the HIMSS EMRAM framework showed a clear divergence between the two institutions (Table 1). DK Hospital had completed foundational stages for laboratory, radiology, and pharmacy integration and had progressed to near complete clinical documentation in ambulatory and emergency services, with gradual extension to inpatient wards. The hospital had initiated medication order entry and partial closed loop processes, positioning it around Stage 3 with movement toward Stage 4–5 and an overall 50% paperless status.

AK Hospital, by contrast, remained concentrated in Stages 1–3, with basic laboratory and pharmacy modules in place and partial clinical documentation in inpatient segments that continued to rely heavily on manual forms. CPOE and advanced clinical decision support functionalities had not yet been implemented, and higher EMRAM stages remained at planning or early design phases. Both hospitals had not achieved Stage 6–7 capabilities related to advanced analytics and full interoperability, although DK Hospital had started planning for enhanced security and analytics features. Overall, DK Hospital exhibited a more advanced and coherent EMR trajectory, whereas AK Hospital operated within a hybrid environment marked by fragmented integration, slower clinical adaptation, and deferred investment in higher stage functionalities.

Comparative SWOT analysis of EMR risk management

The SWOT analysis highlighted contrasting strength and weakness profiles while revealing similar opportunity and threat structures (Table 2). DK Hospital's strengths included progressive system integration across clinical departments, strong institutional commitment to quality improvement, relatively robust access control, and regular staff training that supported evidence based decision making. Its main weaknesses were limited IT personnel, manual risk management reporting using immature applications, persistent (though reduced) system downtime, and the absence of specialised tools for inherent and residual risk reporting.

AK Hospital's strengths centred on the flexibility of its modular development approach, effective cross departmental collaboration during system migration, structured post migration evaluations, and emerging data security training initiatives. However, weaknesses were pronounced in incomplete integration between subsystems, suboptimal laboratory–pharmacy bridging, slow human resource adaptation especially among senior staff, and continued reliance on hybrid documentation.

Both hospitals shared similar opportunity spaces, including national digital transformation momentum, prospects for expanding paperless workflows, and potential adoption of artificial intelligence based decision support. They also faced common threats in the form of human error risks, data inconsistency from hybrid paper–electronic systems, dependence on key IT staff, service disruption during downtime, and regulatory ambiguity regarding the legal status of electronic signatures.

Plan-Do-Check-Act (PDCA) cycle implementation in EMR risk management

Application of the PDCA cycle showed that both hospitals had established basic planning and implementation routines but diverged in monitoring and corrective action maturity (Table 3). In the Plan phase, both DK and AK Hospitals systematically identified and mapped EMR related risks (such as downtime, data breaches, input errors, and

Table 1
EMR risk domains.

Domain	Illustrative issues
Technical/system	Downtime, integration failure, interface instability
Security/privacy	Unauthorised access, data leakage, weak authentication
Human error	Incorrect data entry, low digital skills, resistance to change
Regulatory/ compliance	Legal exposure, accreditation gaps, BPJS and policy misalignment
Operational continuity	Backup failure, service disruption, resource constraints

Table 2
Data collection overview.

Technique	Volume / participants	Focus
Semi-structured interviews	18 participants (leaders, clinicians, IT, records, quality)	EMR use, risk perceptions, governance
FGDs	3 FGDs × 6 participants	Shared experiences and challenges
Non-participant observation	Key clinical units using EMR	Workflows, login, documentation, downtime
Document review	Policies, SOPs, logs, risk reports, security protocols	Formal structures and evidence of practice

Table 3
EMRAM stage assessment results.

EMRAM Stage	DK Hospital	AK Hospital
Stage 0 (Basic System)	Complete	Complete
Stage 1 (Lab, Radiology, Pharmacy)	Complete	Completed for ambulatory & emergency services only
Stage 2 (Clinical Data Repository)	Complete	Partially integrated: inpatient & laboratory
Stage 3 (Clinical Documentation & CDSS)	Nearly all units (ambulatory & emergency complete; inpatient progressive)	Limited segments only (inpatient slow adaptation)
Stage 4 (CPOE Medication)	Early integration implementation initiated	Not yet operational; integration plan incomplete
Stage 5 (Closed Loop Medication)	50% paperless; manual forms remain in inpatient	Migration in progress; hybrid system persists
Stage 6 (Full Physician Documentation)	Planning phase; not fully implemented	Advanced CDSS/ interoperability absent
Stage 7 (Analytics & Interoperability)	Security and analytics planning	Internal security; analytics suboptimal

infrastructure limitations) through internal discussions, end user input, and analysis of historical incidents.

In the Do phase, DK Hospital implemented dual backup systems with daily manual backups at fixed times, dual failover servers, role based access controls, and regular user training, reflecting a focus on redundancy and operational continuity. AK Hospital prioritised fibre optic network infrastructure, independent local servers, and basic access controls to safeguard performance and security, although redundancy mechanisms were less formalised.

The most significant differences emerged in the Check and Act phases. DK Hospital had structured manual reporting, regular monitoring by quality and IT teams, and routine complaint evaluation, supporting more consistent feedback loops and formalised continuous improvement actions through SOP development, application refinement, and tailored training programmes. AK Hospital, in contrast, had only partially implemented monitoring routines, with under developed risk documentation and delayed feedback integration, and while improvement initiatives existed, they lacked the same level of systematic documentation and follow through.

EMR risk taxonomy and comparative vulnerability profiles

Mapping against the predefined five risk domains revealed distinct but overlapping vulnerability patterns across the two hospitals (Table 4).

Table 4
SWOT analysis comparison.

SWOT Dimension	DK Hospital	AK Hospital
Strengths	• Progressive system integration • Institutional commitment to service quality improvement • Robust access control • Regular staff training programs • Integrated systems facilitating clinical information flow • Evidence-based decision-making capability	• Modular approach with customization flexibility • Effective cross-departmental collaboration • Structured post-migration evaluation • Data security training initiatives initiated
Weaknesses	• Limited information technology personnel capacity • Manual risk management reporting processes • Persistent system downtime incidents despite frequency reduction • Immature risk management applications • Absence of specialized tools for inherent and residual risk reporting • Documentation using Microsoft Excel	• Incomplete comprehensive system integration • Suboptimal laboratory-pharmacy bridging • Slow human resource adaptation • Challenges with senior staff accustomed to legacy systems
Opportunities	• Paperless hospital digitalization expansion • System development with customization • Clinical service process efficiency improvement • National digital transformation momentum (government regulation) • AI and machine learning adoption for clinical decision support enhancement	• Paperless hospital digitalization expansion • System development with customization • Clinical service process efficiency improvement • National digital transformation momentum (government regulation) • AI and machine learning adoption for clinical decision support enhancement
Threats	• Data input human error risk • Hybrid manual-electronic systems creating data inconsistency potential • Key IT personnel dependency • System downtime disrupting service continuity • Electronic signature (TTE) regulatory ambiguity	• Data input human error risk • Hybrid manual-electronic systems creating data inconsistency potential • Key IT personnel dependency • System downtime disrupting service continuity • Electronic signature (TTE) regulatory ambiguity

1. Technical and system risks: DK Hospital had substantially reduced system downtime compared with earlier phases, with recent incidents typically lasting 15–20 min, although a prior six hour outage illustrated residual vulnerability. Ongoing issues included uncoordinated construction causing network cable disruptions and separation between billing and EMR systems that required dual data entry for some transactions. AK Hospital faced more fundamental technical constraints: multi vendor database fragmentation, incomplete bridging between pharmacy and laboratory modules, and a complex migration to a central PostgreSQL database that affected workflow efficiency and medication safety oversight.
2. Security and privacy risks: both hospitals operated largely offline systems with restricted external access, supported by password based access control and role based authorisation schemes. DK Hospital supplemented these measures with locked server rooms, licensed antivirus software, and firewall configurations, but relied on ad hoc security incident documentation without a dedicated EMR incident tracking system. AK Hospital combined independent local servers and fibre optic infrastructure to minimise cyber exposure, yet security documentation was minimal and the last formal assessment had occurred during the 2023 accreditation process.
3. Regulatory and compliance risks: both hospitals integrated with the national BPJS system and thus depended on external system stability; national BPJS downtime directly disrupted services despite internal controls. Regulatory ambiguity regarding the legal validity of electronic signatures persisted in both settings, creating medicolegal uncertainty around fully paperless workflows. AK Hospital additionally faced challenges in meeting integration requirements for inpatient pharmacy services due to incomplete bridging with laboratory and clinical systems.
4. Operational continuity risks: DK Hospital had established dual backup servers, daily manual backups, and ring topology network redundancy to support service continuity, though transactions occurring after backups remained at risk of data loss during major failures. In AK Hospital, EMR utilisation in the emergency department remained problematic; high patient flow before registration forced staff to maintain parallel paper documentation, undermining full digital transition in one of the most critical clinical

Table 5
PDCA implementation assessment.

PDCA Phase	DK Hospital	AK Hospital
Plan (Planning)	• Risk identification and mapping (system downtime, data breach, input error, infrastructure limitation) • Internal risk management team discussions • End-user system input • Historical incident analysis	• Risk identification and mapping (system downtime, data breach, input error, infrastructure limitation) • Internal risk management team discussions • End-user system input • Historical incident analysis
Do (Implementation)	• Dual data backup systems • Daily manual backup at 14:00 • Dual failover servers for system failure prevention • Password-based access control • Role-based access restrictions • Regular periodic user training programs	• Fiber optic-based network infrastructure • Independent local servers • Password-based access control • Role-based access restrictions • Regular periodic user training programs • Focus on security and access speed
Check (Monitoring)	• Structured manual reporting systems • Regular monitoring by quality and IT teams • System complaint evaluation • Semester periodic reporting	• Risk documentation systems under development • Partial monitoring and feedback implementation • Monitoring acceleration required • Risk documentation not yet comprehensive
Act (Action)	• Structured SOP development • Application development aligned with requirements • Customized training programs • Module adjustment based on evaluation • Consistent continuous improvement commitment	• SOP development • Application development aligned with requirements • Customized training programs • Module adjustment based on evaluation • Continuous improvement requiring acceleration

Table 6
EMR risk taxonomy and comparative vulnerability profiles.

Risk Category	Risk Type	DK Hospital	AK Hospital
Technical & System Risks	System Downtime	Downtime significantly reduced; current duration 15–20 min; most recent incident 6 h (23:00–05:00)	Complex system integration; fragmented database from multiple vendors
	Network Disruptions	Disruptions caused by uncoordinated construction; network cable disconnection	Not identified as primary issue
	System Integration	Billing and EMR systems separated; dual data entry required for some data	Fragmented database; PostgreSQL centralization migration ongoing
Security & Privacy Risks	Access Control	Username and password protocols; locked secured server rooms; licensed antivirus; Mikrotik-configured firewall	Independent local servers; fiber optic infrastructure; layered access schemes
	Cyber Security	Offline system; inaccessible from external networks; minimal cyber risk	Offline system; independent local servers; cyber risk minimized
	Incident Documentation	Ad-hoc documentation without systematic EMR incident tracking	Minimal documentation; last formal assessment during 2023 accreditation
Regulatory & Compliance Risks	BPJS (Social and Health Insurance) Compliance	Integrated BPJS system; national BPJS downtime causes service disruption	Integrated BPJS system; adherence to BPJS policy (one service per patient daily)
	Electronic Signature Regulation	Gray area; legal validity concerns for electronic signature (TTE) unclear	Gray area; legal validity concerns for electronic signature (TTE) unclear
	External System Integration	BPJS integration operational; external system dependency remains	Laboratory-pharmacy integration suboptimal; inpatient pharmacy incomplete bridging
Operational Risks	Service Continuity & Backup	Dual backup systems (2 failover servers); daily manual backup 14:00; ring topology redundancy	Emergency department EMR effectiveness challenged; chaotic patient flow; paper documentation persistence required
	Emergency Department Operations	Not specifically analyzed	Emergency department EMR challenged; patients arrive prior to system registration
	Key Personnel Dependency	7 IT staff operating single shift plus on-call; burnout risk documented	Reliance on key programmer personnel; transition risk
Human Resource Risks	IT Capacity Limitations	7 IT personnel; ideal requirement 15 personnel; recruitment barriers via P3K/CPNS pathways	High vendor and key programmer dependency
	Vendor/Programmer Dependency	Minimal; internal system development	Internal system development reduces external vendor dependency
	Data Input Human Error	EMR system enforces field completion; door-to-door training during implementation	Patient data input errors; double entry for patients inactive $>$ 5 years
	Patient Data Duplication	No significant issues documented	Legacy patients reregistered as new patients causing data duplication
	Slow Human Resource Adaptation	Periodic retraining required due to internship physician rotation (6-month cycles)	Slow SDM adaptation; senior staff difficulty; ongoing intensive coaching required

environments. Both hospitals reported that emergency workflows remained a stress point for EMR resilience, albeit through different mechanisms.

5. Human resource risks: DK Hospital operated EMR and related systems with seven IT staff, below the internally estimated requirement of 15, resulting in single shift coverage with on call support and documented burnout risk. The hospital mitigated data entry errors through mandatory field completion in the EMR interface and intensive door to door training during implementation, but frequent rotation of internship physicians required recurring retraining cycles. AK Hospital showed stronger dependence on a small number of key programmers, raising continuity risks if these individuals became unavailable. Additional challenges included patient data duplication (especially when inactive patients were re registered as new) and slower adaptation among senior staff accustomed to paper based routines, necessitating ongoing intensive coaching.

Taken together, these findings indicate that DK Hospital has

Table 7
Strategic priorities for EMR risk governance.

Level / actor	Primary focus	Concise recommendations
DK Hospital	Consolidating higher maturity	Complete full inpatient digitalisation and reduce hybrid documentation; Replace spreadsheet-based risk logs with an integrated dashboard that combines incidents, downtime, and compliance indicators; Strengthen cyber-security and interoperability as prerequisites for higher EMRAM stages, including a documented disaster-recovery plan.
	IT workforce capacity	Align IT staffing levels with digital workload and planned expansion; use risk and continuity arguments in negotiations with hospital management and local government.
AK Hospital	Consolidating EMR foundations and integration	Accelerate retraining for all staff, with targeted support for senior clinicians who are least familiar with EMR; Complete EMR–pharmacy–laboratory integration for inpatient services to support medication safety and BPJS claims; Establish clear downtime-management procedures in the emergency department to reduce dependence on paper.
	Reducing key-person dependency	Develop succession and cross-training plans for core EMR developers; document system architecture and recovery procedures to avoid reliance on single individuals.
Both hospitals	Strengthening integrated governance	Adopt a unified governance framework that combines EMRAM stages with ISO 31000 and COBIT 5 to prioritise risks, select mitigation actions, and track progress; Implement EMR risk dashboards that display key indicators (incidents, downtime duration, data-entry compliance, security events) in near real time; Establish cross-functional rapid-response teams (IT, clinical, quality) for operational and cyber incidents.
Policymakers / hospital networks	System-level resilience based on EMR	Recognise EMR risk-management capacity as a core dimension of health-system resilience, alongside clinical quality and financial sustainability; Promote mentoring and benchmarking arrangements in which more mature hospitals, such as DK, support peers with lower digital-governance capacity; Embed EMR risk indicators into accreditation standards and routine reporting requirements.

achieved comparatively higher EMR maturity and more institutionalised risk management routines, albeit under human resource and infrastructure constraints, whereas AK Hospital exhibits greater flexibility through modular development but remains more exposed to integration failures, documentation gaps, and key person dependency. These empirically observed patterns provide the basis for the subsequent discussion on uncertainty, distributed control, and diagnostic analytics in EMR risk governance.

Discussion

Limitations of existing studies and rationale for the proposed governance roadmap

Prior EMR research in Indonesia and comparable settings has concentrated on adoption status, user satisfaction, perceived benefits, and technical challenges, often treating risk management as a secondary concern. [15,19,17] As a result, key dimensions such as formal risk identification routines, incident documentation systems, and integration of EMR risk into enterprise wide governance remain under theorised and weakly measured. [23,29] Comparative analyses across ownership types and accreditation statuses are also scarce, limiting insight into how institutional characteristics shape EMR risk management trajectories. [16,29,18]

Hospital risk management literature, in turn, tends to address clinical risk, occupational safety, or accreditation related processes at an aggregate level, without separating out risks arising from digital infrastructures such as EMR.[3,21,22] Studies often report generic barriers (limited staff engagement, incomplete reporting, and weak feedback loops) without specifying how these issues manifest in digital health systems or how they intersect with BPJS integration, cyber security requirements, and medico legal uncertainty. [2,3,14,21]

By contrast, microgrid and transformer diagnostics research explicitly models uncertainty, distributed control, and high dimensional diagnostics as core design problems and uses these models to derive operational strategies, resilience metrics, and decision rules.[24,26,27] This contrast exposes a theoretical and practical gap: EMR risk governance has not yet fully internalised uncertainty oriented, data driven frameworks that are already common in other safety critical infrastructures.[9]

The recommendations advanced in this study (integrating EMRAM maturity assessments with ISO 31000 and COBIT 5, developing structured EMR risk dashboards, and institutionalising PDCA based learning cycles) are intended to address these gaps. They translate the logic of distributed control and advanced diagnostics into the hospital EMR context while remaining grounded in empirical observation of DK and AK Hospitals. This alignment between conceptual advances in uncertainty theory and situated empirical evidence provides a strong justification for the proposed governance roadmap.[26]

Uncertainty, distributed control, and diagnostic analytics in EMR risk governance

The comparative findings from Kudus can be interpreted through the lens of uncertainty management in complex, distributed infrastructures. Research on islanded microgrids shows that stability under variable renewable generation depends not only on robust hardware but also on decentralised control architectures that coordinate distributed generation, storage, and load shedding using local information and adaptive rules.[25] Smart scheduling studies further demonstrate that integrated optimisation of power management, voltage control, and distributed solutions is necessary to balance cost, reliability, and resilience under prediction errors and external shocks.[24]

An analogous pattern appears in EMR risk management maturity. DK Hospital's higher EMRAM stage, routine PDCA cycles, and structured risk reporting resemble a distributed control system in which clinical units, IT, quality, and management jointly detect, report, and mitigate EMR related risks. By contrast, AK Hospital's fragmented documentation, hybrid paper–electronic workflows, and dependence on a small number of key IT staff parallel a centralised, brittle control architecture that is less able to absorb uncertainty or learn from incidents.

From a diagnostic analytics perspective, transformer winding studies show how uncertainty in complex signals can be handled through hybrid models that combine discrete wavelet transforms, logistic regression, and neural networks to classify subtle fault signatures.[26] Complementary frequency response work uses hyperbolic fuzzy cross entropy to interpret ambiguous responses and distinguish healthy from faulty states across frequency bands.[27] These approaches share two features: they treat uncertainty as a quantifiable property of high dimensional data and they embed diagnostics in routine monitoring workflows.[26,27]

In EMR ecosystems, incident logs, access records, error reports, and downtime data constitute a comparable diagnostic layer that, if systematically captured and analysed, could support early detection of risk hotspots and pre emptive interventions.[3,4,9] The present study indicates

that both hospitals collect some incident and performance data, but these data are not yet organised into continuous, analytically rich diagnostic systems akin to advanced power systems. Embedding uncertainty aware analytics into EMR risk dashboards such as anomaly detection in login patterns, clustering of error types, and trend analysis of downtime would operationalise this theoretical lens and move hospitals toward intelligent, learning EMR risk management systems.[3,15,26]

Conclusion

This comparative qualitative study shows that EMR implementation in Kudus hospitals is best understood as an organizational transformation process in which technical, human, regulatory, and operational risks evolve together, rather than as a discrete IT project. DK Hospital, which is progressing toward EMRAM Stage 5 with approximately half of its documentation already paperless, has begun to embed EMR risk within structured PDCA cycles, formal risk reporting systems, and continuous training programs, thereby developing a more resilient digital infrastructure. AK Hospital, by contrast, is still consolidating foundational EMR capabilities and faces persistent challenges related to system integration, incomplete incident documentation, and high dependency on a small number of IT personnel.

Across both institutions, five recurring vulnerability domains were identified: (1) technical risks related to downtime, fragmented databases, and incomplete system bridging; (2) security and privacy risks stemming from hybrid architectures and limited incident logging; (3) human error risks linked to data entry mistakes, inconsistent adoption across staff cohorts, and legacy paper habits; (4) regulatory and compliance risks associated with BPJS integration and ambiguity around electronic signatures; and (5) operational continuity risks driven by limited IT staffing, single developer dependency, and emergency department workflow misalignment with EMR processes. These vulnerabilities mirror patterns documented in other critical infrastructures, where uncertainty, decentralization, and complex interactions between subsystems require continuous monitoring and adaptive control rather than static protocols.

The study contributes to the literature by integrating HIMSS EMRAM maturity assessment, PDCA cycle analysis, and SWOT mapping into a single evaluative framework for EMR risk governance in developing health systems. This multi lens approach highlights that higher EMRAM stages are associated not merely with more advanced software modules, but with deeper institutionalization of risk management routines, stronger cross unit coordination, and greater capacity to learn from incidents. Conversely, hospitals that treat EMR primarily as a technical upgrade, without parallel investment in governance, documentation, and human resource development, remain vulnerable despite digitalization.

Building on insights from uncertainty theory and diagnostic analytics in microgrids and transformer systems, the paper argues that EMR risk governance should evolve toward an integrated, data driven architecture in which risk indicators are continuously captured, analyzed, and acted upon. Concretely, this implies coupling EMRAM based roadmaps with ISO 31000 and COBIT 5 frameworks, replacing spreadsheet based risk logs with structured dashboards, and deploying advanced analytics to detect anomalies in EMR usage, access, and performance. For DK Hospital, strategic priorities include completing full inpatient digitalization, expanding IT staffing, and strengthening cyber security and interoperability to prepare for higher EMRAM stages. For AK Hospital, priorities include accelerating staff retraining, completing system integration (particularly pharmacy and laboratory bridging), institutionalizing risk documentation, and reducing single developer dependency through cross training and succession planning.

More broadly, the findings suggest that national policymakers and hospital networks should treat EMR risk management capacity as a core dimension of health system resilience, on par with clinical quality and financial sustainability. Future research could build on this study by combining qualitative comparative analyses with quantitative risk and resilience metrics, and by piloting uncertainty aware diagnostic tools for EMR incident data analogous to those used in advanced power systems monitoring. Such work would further close the gap between conceptual advances in uncertainty theory and the practical governance of digital health infrastructures in low and middle income countries

Implications for framework development

The findings indicate that EMR risk-management maturity is primarily an institutional capability outcome rather than a simple function of technology acquisition. Redundant servers or closed-loop modules are insufficient without parallel investment in human resources, governance structures, and formal procedures. DK Hospital illustrates that progression toward EMRAM Stage 5 is driven by internal ownership of governance, phased change management, and stakeholder alignment, even under fiscal constraints similar to other public hospitals. This stands in contrast to broader national patterns in which only a subset of regional hospitals implement risk management comprehensively, leaving wide variation in systemic capacity.

A central gap lies between the ability to map risks and the ability to monitor and learn from them consistently. Both hospitals can identify critical risks, but conversion of risk registers into regular monitoring and documented learning cycles remains weak, particularly in AK Hospital. This pattern mirrors other resource-constrained health systems where risk procedures are codified on paper yet responses remain reactive and incident-driven.

These observations imply the need for an EMR risk-governance framework that emphasises three levers: digitalisation of risk documentation, partial automation of monitoring, and scheduled incident reviews. Such a framework would prevent risk mapping from becoming a static list and instead embed it in a functioning PDCA loop aligned with established standards such as ISO 31000 and COBIT 5.[30–32]

Strategic implications and recommendations

The study underscores that sustainable EMR risk governance requires a shift from sporadic incident response to proactive stewardship of a socio-technical system. Integrating EMRAM, ISO 31000, COBIT 5, and PDCA can provide a shared scaffold for predicting, preventing, and responding to system failures in a structured way.

Overall, the strategic implications point to the need for a measurable, iterative, and data-driven architecture for EMR risk governance, rather than one-off IT projects or document-centred compliance. By binding EMRAM, ISO 31000, COBIT 5, and PDCA into a single, context-sensitive framework, hospitals can move from partial digitalisation toward auditable and accountable digital resilience.

Ethics statement

This research received ethical approval from the Institutional Review Board (IRB) of DK Hospital for both hospitals (approval numbers: [67/KEPK/RSLH/VI/2025] issued on August 16 2025). The study was conducted in strict adherence to Indonesian research ethics guidelines and the principles outlined in the Declaration of Helsinki (2013). All participants provided written informed consent prior to data collection. Participant autonomy was upheld through clear disclosure of research objectives, the voluntary nature of participation, and the right to withdraw without consequence.

All participant identities were de-identified during data transcription and reporting to ensure confidentiality. Research procedures involved only interviews, focus group discussions, observation, and document review. Research activities classified as minimal risk since they did not involve clinical intervention, medical procedures, or manipulation of clinical data. Participant confidentiality was maintained through secure data storage, password-protected servers, and restricted access limited

to the research team.

No personal identifying information (names, hospital ID numbers, contact details) was retained in final research documents. All raw data and recordings are stored securely and will be retained for five years post-publication, after which they will be securely destroyed in accordance with data protection regulations.

CRediT authorship contribution statement

Jeki Pornomo: Writing – review & editing, Writing – original draft, Visualization, Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **Cikita Berlian Hakim:** Visualization, Validation, Supervision, Software, Resources, Project administration, Investigation, Funding acquisition. **Ingrid Novi Aryani:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **Taufiqi Dwi Wijayanti:** Visualization, Validation, Supervision, Software, Resources, Project administration, Investigation, Funding acquisition.

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Conflict of Interest (COI) Statement

The authors declare no conflicts of interest related to this research. Neither author has received financial support, consultancy fees, employment benefits, stock holdings, or other material gains from organizations that may be affected by the findings of this study.

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